

Module 2 : Syntax Analysis - Morphological Analysis

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Agenda

- Syntax Analysis - Introduction
- POS tagging
- Tagset for English
- Generative Models
- HMM
- CRF
- Parsers

Syntax Analysis - Introduction

- Syntax analysis or Parsing is the **important phase of NLP**. The purpose of this phase is to draw **exact meaning or dictionary meaning from the text**.
- In NLP, Syntactic analysis is used to **assess how the natural language aligns with the grammatical rules**.
- Computer algorithms are used to **apply grammatical rules to a group of words** and derive meaning from them.
- Syntactic analysis helps us **understand the roles played by different words** in a body of text.
- Consider **“Innocent peacefully children sleep little”** vs **“Innocent little children sleep peacefully”**

Syntax Analysis - Introduction

- **A basic set:**
 - N, V, Adj, Adv, Prep, Det, Aux, Part, Conj, Num
- **A simple division:** open/content vs. closed/function
 - Open: N, V, Adj, Adv
 - Closed: Prep, Det, Aux, Part, Conj, Num
- **Many subclasses, e.g.**
 - eats/V -> eat/VB, eat/VBP, eats/VBZ, ate/VBD, eaten/VBN, eating/VBG, ..
 - Reflect morphological form & syntactic function

Closed classes in English

- **Prepositions:** on, under, over, near, by, at, from, to, with
- **Particles:** up, down, on, off, in, out, at, by
- **Determiners:** a, an, the
- **Conjunctions:** and, but, or, as, if, when
- **Pronouns:** she, who, I, others
- **Auxiliary verbs:** can, may, should, are
- **Numerals:** one, two, three, first, second, third

Parsing..

Given a sentence and a grammar:

- Checks that **the sentence is correct according with the grammar &**
- if so returns **a parse tree representing the structure of the sentence**

Challenges: Ambiguity in the Language

- **Ex. : I made her duck.**
- How many **different interpretations** does this sentence have?
- What are the **reasons for the ambiguity?**
- The **categories of knowledge of language** can be thought of as ambiguity resolving components.
- How can **each ambiguous piece be resolved?**

Challenges: Ambiguity in the Language

Some interpretations of: **I made her duck.**

1. I **cooked *duck*** for her.
2. I **cooked *duck*** belonging to her.
3. I **created a toy duck** which she owns.
4. I caused her to **quickly lower her head** or body.
5. I used magic and **turned her into a *duck***.

Challenges: Ambiguity in the Language

- duck – morphologically and syntactically ambiguous:
- Noun or Verb
- her – syntactically ambiguous: dative or possessive.
- make – semantically ambiguous: cook or create.
- make – syntactically ambiguous:
- Transitive – takes a direct object. => 2
- Di-transitive – takes two objects. => 5
- Takes a direct object and a verb. => 4

Syntax Analysis : Terms & Techniques

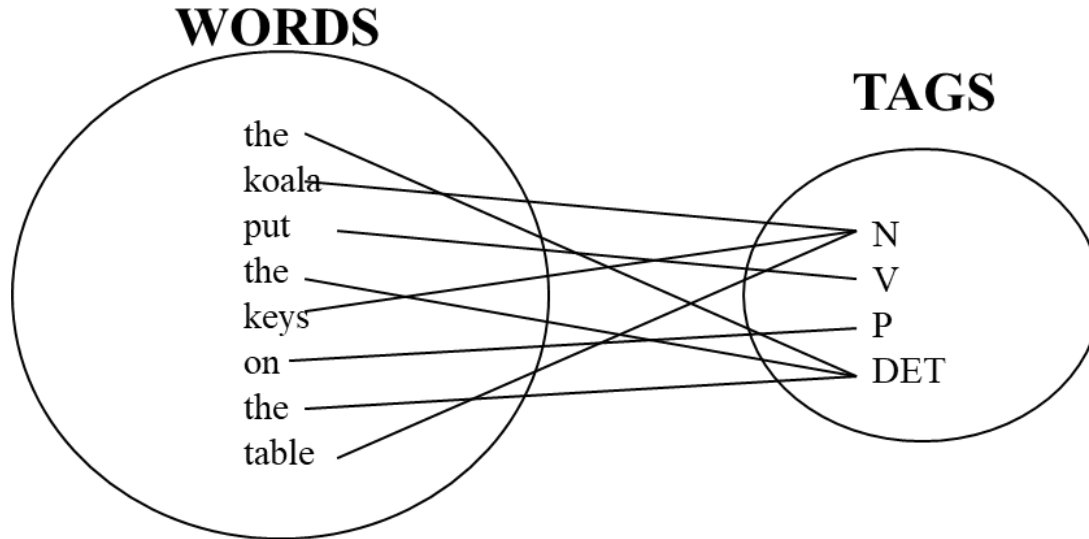
- **Lemmatization** : entails reducing the various inflected forms of a word into a single form for easy analysis
- **Morphological Segmentation** : Involves dividing words into individual units called morphemes
- **Word Segmentation** : Involves dividing a large piece of continuous text into distinct units
- **POS Tagging** : Involves **identifying the part of speech for every word**
- **Sentence breaking** : Involves placing sentence boundaries on a large piece of text
- **Stemming** : Involves cutting the inflected words to their root form.

Part of Speech (POS) Tagging

- Is the task of **assigning POS tags to words**
- **Selecting** among more than one tags that apply
- Can be used for further **NLP tasks**
- **Use Cases:** **Information extraction, Question Answering** etc.

What is Part of Speech (POS) Tagging?

- The process of **assigning a part-of-speech** or lexical class marker **to each word in a corpus**



Part of Speech (POS) Tagging?

- It is a process that **attaches each word in a sentence with a suitable tag** from a given set of tags.
- The set of tags are called as **Tag-set**.
- **Standard Tag-set : Penn Tree (for English)**

Part of Speech (POS) Tagging?

- It is a process of **converting a sentence to forms** – list of words, list of tuples (where each tuple is having a form (word, tag)).
- The **tag in case of it is a part of speech tag**, and signifies whether the word is a noun, adjective, verb, and so on.
- Automatic **assignment of descriptors to the given tokens** is called **Tagging**.
- The descriptor is called tag.
- **Book/VB that/DT flight/NN**

Part of Speech (POS) Tagset

- Brown tagset (87 tags)
 - Brown corpus
- C5 tagset (61 tags)
- C7 tagset (146 tags!)
- Penn TreeBank (45 tags)
 - A large annotated corpus of English tagset

How do we choose the tagset?

- <http://www.comp.leeds.ac.uk/amalgam/tagsets/tagmenu.html>
- Brown Corpus (Francis & Kucera '82), 1M words, 87 tags.
- Penn Treebank: hand-annotated corpus of Wall Street Journal, 1M words, 45-46 tags

Part of Speech (POS) Tagging

- POS Taggers try to find POS tags for the words.
- duck is a verb or noun? (morphological analyzer cannot make decision).
- A POS tagger may make that decision by looking the surrounding words.
- Duck! (verb)
- Duck is delicious for dinner. (noun)

Penn Treebank Tagset

- The Penn Treebank tagset was selected from the original 87-tags tagset for the Brown corpus
- The Penn Treebank tagset, has been applied to the Brown corpus and a number of other corpora.
- Ex: The/DT grand/JJ jury/NN commented/VBD on/IN a/DT number/NN of/IN other/JJ topics/NNS ./.



Number	Tag	Description
1.	CC	Coordinating conjunction
2.	CD	Cardinal number
3.	DT	Determiner
4.	EX	Existential <i>there</i>
5.	FW	Foreign word
6.	IN	Preposition or subordinating conjunction
7.	JJ	Adjective
8.	JJR	Adjective, comparative
9.	JJS	Adjective, superlative
10.	LS	List item marker
11.	MD	Modal
12.	NN	Noun, singular or mass
13.	NNS	Noun, plural
14.	NNP	Proper noun, singular
15.	NNP S	Proper noun, plural
16.	PDT	Predeterminer

Number	Tag	Description
17.	POS	Possessive ending
18.	PRP	Personal pronoun
19.	PRPS	Possessive pronoun
20.	RB	Adverb
21.	RBR	Adverb, comparative
22.	RBS	Adverb, superlative
23.	RP	Particle
24.	SYM	Symbol
25.	TO	<i>to</i>
26.	UH	Interjection
27.	VB	Verb, base form
28.	VBD	Verb, past tense
29.	VBG	Verb, gerund or present participle
30.	VCN	Verb, past participle
31.	VBP	Verb, non-3rd person singular present
32.	VBZ	Verb, 3rd person singular present
33.	WDT	Wh-determiner
34.	WP	Wh-pronoun
35.	WPS	Possessive wh-pronoun
36.	WRB	Wh-adverb

Penn Treebank Tagset

Tag	Description	Example	Tag	Description	Example	Tag	Description	Example
CC	coordinating conjunction	<i>and, but, or</i>	PDT	predeterminer	<i>all, both</i>	VBP	verb non-3sg present	<i>eat</i>
CD	cardinal number	<i>one, two</i>	POS	possessive ending	<i>'s</i>	VBZ	verb 3sg pres	<i>eats</i>
DT	determiner	<i>a, the</i>	PRP	personal pronoun	<i>I, you, he</i>	WDT	wh-determ.	<i>which, that</i>
EX	existential 'there'	<i>there</i>	PRP\$	possess. pronoun	<i>your, one's</i>	WP	wh-pronoun	<i>what, who</i>
FW	foreign word	<i>mea culpa</i>	RB	adverb	<i>quickly</i>	WP\$	wh-possess.	<i>whose</i>
IN	preposition/ subordin-conj	<i>of, in, by</i>	RBR	comparative adverb	<i>faster</i>	WRB	wh-adverb	<i>how, where</i>
JJ	adjective	<i>yellow</i>	RBS	superlatv. adverb	<i>fastest</i>	\$	dollar sign	<i>\$</i>
JJR	comparative adj	<i>bigger</i>	RP	particle	<i>up, off</i>	#	pound sign	<i>#</i>
JJS	superlative adj	<i>wildest</i>	SYM	symbol	<i>+, %, &</i>	"	left quote	<i>' or "</i>
LS	list item marker	<i>1, 2, One</i>	TO	"to"	<i>to</i>	"	right quote	<i>' or "</i>
MD	modal	<i>can, should</i>	UH	interjection	<i>ah, oops</i>	(	left paren	<i>[, (, {, <</i>
NN	sing or mass noun	<i>llama</i>	VB	verb base form	<i>eat</i>	)	right paren	<i>],), }, ></i>
NNS	noun, plural	<i>llamas</i>	VBD	verb past tense	<i>ate</i>	,	comma	<i>,</i>
NNP	proper noun, sing.	<i>IBM</i>	VBG	verb gerund	<i>eating</i>	.	sent-end punc	<i>. ! ?</i>
NNPS	proper noun, plu.	<i>Carolinas</i>	VBN	verb past part.	<i>eaten</i>	:	sent-mid punc	<i>: ; ... --</i>

Figure 8.1 Penn Treebank part-of-speech tags (including punctuation).



How do we assign POS tags to words in sentence?

Get/V the/Det bass/N

Exercise :

Time flies like an arrow.

Fruit flies like a banana.

The flies like a banana

How do we assign POS tags to words in sentence?

- Time/[V,N] flies/[V,N] like/[V,Prep] an/Det arrow/N
- Time/N flies/V like/Prep an/Det arrow/N
- Fruit/N flies/N like/V a/DET banana/N
- Fruit/N flies/V like/V a/DET banana/N
- The/Det flies/N like/V a/DET banana/N

For English or Hindi...

- I **bank** on the **bank** on the river **bank** for my transactions [English]

noun

verb

verb

- Mujhe **khaanna** **khaanna** hai .[Hindi]

Noun

verb

POS Tag Ambiguity

- I **bank** on the **bank** on the river **bank** for my transactions [English]

noun

verb

verb

- Mujhe **khaanna** **khaanna** hai .[Hindi]

Noun

verb

- पुजा** देवीची **पुजा** कर. [Marathi]

Noun

verb

Why is Tagging Hard?

- Example
 - Book/VB that/DT flight/NN
 - Does/VBZ that/DT flight/NN serve/VB dinner/NN
- Tagging is a type of disambiguation
- Book can be NN or VB
- Can I read a book on this flight?
- That can be a DT or complementizer ?
- My travel agent said that there would be a meal on this flight.

The Problem is..

- Words often have more than one word class: **this**
- **This** is a nice day = PRP
- **This** day is nice = DT
- You can go **this** far = RB (Adverb)

POS Ambiguity

Types:		WSJ	Brown
Unambiguous	(1 tag)	44,432 (86%)	45,799 (85%)
Ambiguous	(2+ tags)	7,025 (14%)	8,050 (15%)
Tokens:			
Unambiguous	(1 tag)	577,421 (45%)	384,349 (33%)
Ambiguous	(2+ tags)	711,780 (55%)	786,646 (67%)

Figure 8.2 Tag ambiguity for word types in Brown and WSJ, using Treebank-3 (45-tag) tagging. Punctuation were treated as words, and words were kept in their original case.

POS Ambiguity

		87-tag Original Brown	45-tag Treebank Brown
Unambiguous (1 tag)		44,019	38,857
Ambiguous (2–7 tags)		5,490	8844
Details:	2 tags	4,967	6,731
	3 tags	411	1621
	4 tags	91	357
	5 tags	17	90
	6 tags	2 (<i>well, beat</i>)	32
	7 tags	2 (<i>still, down</i>)	6 (<i>well, set, round, open, fit, down</i>)
	8 tags		4 (<i>'s, half, back, a</i>)
	9 tags		3 (<i>that, more, in</i>)

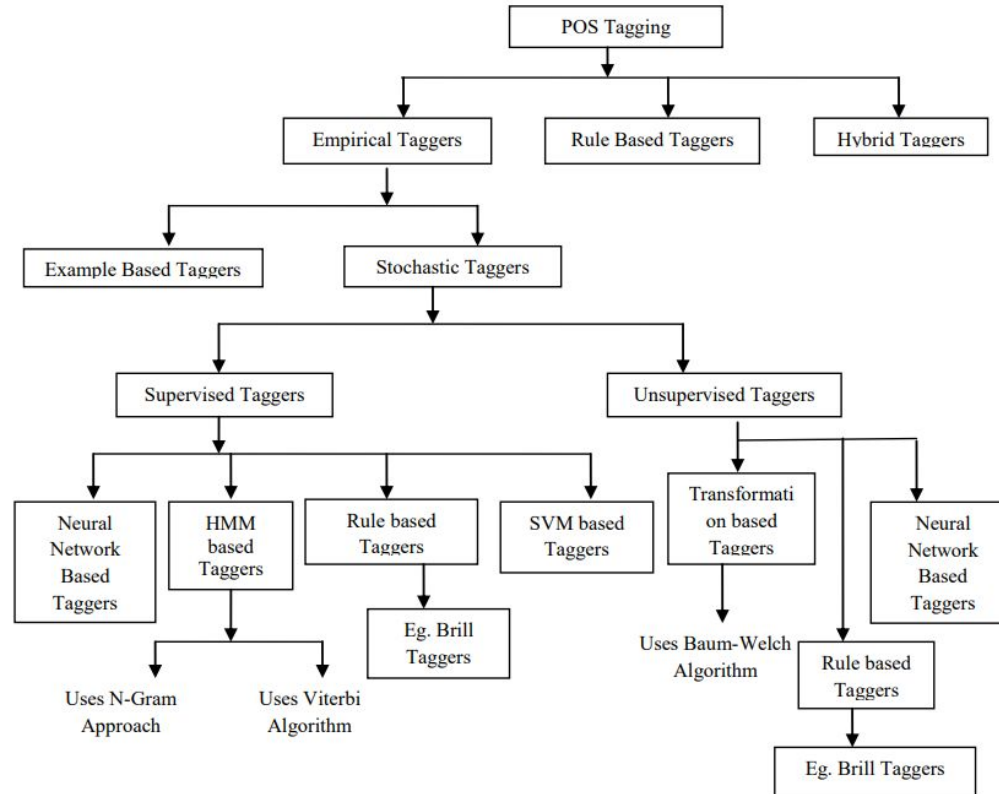
How to Solve? Baseline Method..

- Tagging **unambiguous word** with the **correct label**
- Tagging **ambiguous word** with their **most frequent label**
- Tagging **unknown words** as a **Noun**
- This method performs around **90% precision**

Classification of POS tagging approaches

- Rule based POS tagging
- Stochastic POS tagging
- Transformation based tagging

Classification of POS tagging approaches



Rule based POS tagging

- One of the **oldest techniques** of tagging
- Rule-based taggers use **dictionary or lexicon** for getting possible tags for tagging each word.
- If the word has more than one possible tag, then rule-based taggers **use hand-written rules** to identify the correct tag.
- **Disambiguation can also be performed** in rule-based tagging by **analyzing the linguistic features of a word** along with its preceding as well as following words.
- For **example**, suppose if the **preceding word of a word is article then word must be a noun.**

Rule based POS tagging

- Standard approach (two steps):
 1. Dictionaries to assign a list of potential tags
 - Plays (NNS/VBZ)
 - well (UH/JJ/NN/RB)
 - with (IN)
 - others (NNS)
 2. Hand-written rules to restrict to a POS tag
 - Plays (VBZ)
 - well (RB)
 - with (IN)
 - others (NNS)

Rule based POS tagging

- **Step 1:** Start with a **dictionary**
- **Step 2 :** **Assign all possible tags to words** from the dictionary
- **Step 3 :** Write **rules by hand** to selectively remove tags
- **Step 4 :** **Leaving the correct tag** for each word.

Step 1 : Start with a dictionary

Input : She promised to back the bill

- she : PRP
- promised : VBN,VBD
- to : TO
- back : VB, JJ, RB, NN
- the : DT
- bill : NN, VB

Step 2 : Write rules to eliminate tags

Eliminate VBN if VBD is an option when VBN|VBD follows “<start> PRP”

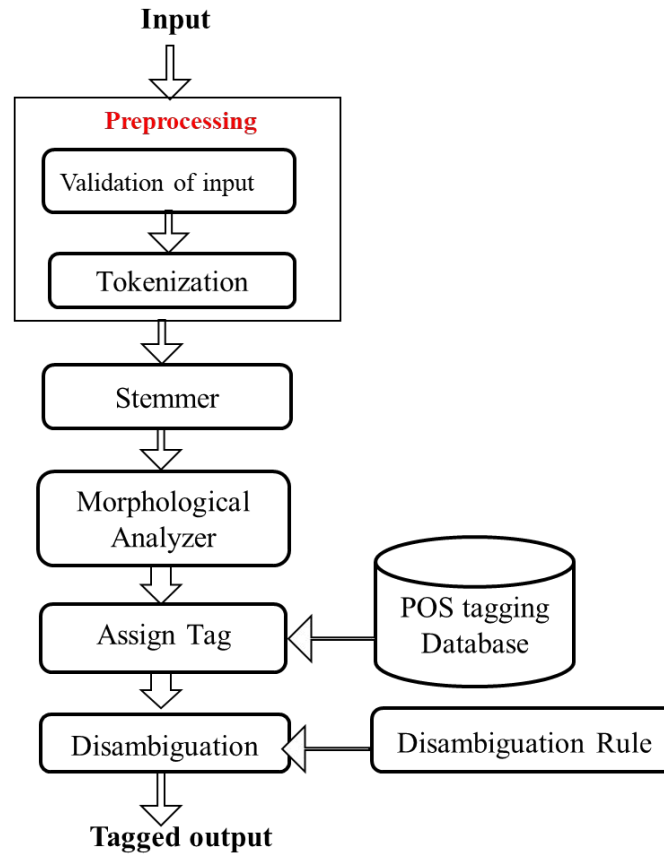
			NN		
			RB		
	VBN		JJ		VB
PRP	VBD	TO	VB	DT	NN

She promised to back the bill

Properties of Rule-Based POS Tagging

Rule-based POS taggers possess the following properties –

- These taggers are **knowledge-driven taggers**.
- The rules in Rule-based POS tagging are **built manually**.
- The information is **coded in the form of rules**.
- We have some limited number of **rules approximately around 1000**.
- **Smoothing and language modeling** is defined explicitly in rule-based taggers.



Marathi Rule based POS tagger System

Rule-Based POS Tagger for Marathi Language

- **Rule based POS tagger**, where each word will be assigned with corresponding tag and if word is assigned with multiple tags then such ambiguity will be removed by Word Sense Disambiguation (WSD) rules.
- **Input:** Accepts a sentence or document in MARATHI .
- **Output:** PoS tagged Document

References:

- [Part of Speech Tagger for Marathi Language \(ijcaonline.org\)](http://ijcaonline.org)
- [Resource Centre for Indian Language Technology Solutions\(CFILT\) \(iitb.ac.in\)](http://iitb.ac.in)
- [GitHub - avineshpvs/indic_tagger: Indian Language Tagger and Chunker \(Hindi, Telugu, Tamil, Marathi, Punjabi, Kanada, Malayalam, Urdu, Bengali\)](https://github.com/avineshpvs/indic_tagger)
- [Part-of-Speech Tagger for Marathi Language using Limited Training Corpora \(ijcaonline.org\)](http://ijcaonline.org)

Validation of Input

- **Input:** Input document
- **Output:** Devanagari script

Reference Paper: [Link](#)

Algorithm

1. Use the character set as sparknlp.org (for marathi language)
 2. Scan the input document.
 3. Compare each character from scanned input document with [SparkNLP](#).
 4. If character is present in the UTF-8, then it is valid to Devanagari script, otherwise not.
 5. Ignore all the invalid Devanagari script characters.
 6. Repeat step 3 till all characters from input script document get verified.
 7. Store all the valid Devanagari character, words in file to process further.
- **Example:** Consider the input sentence as:
 - **Input:** मराठी भाषा हे महाराष्ट्राचे वैभव आहे. It is given in history of Marathi Language
 - **Output:** मराठी भाषा हे महाराष्ट्राचे वैभव आहे.

Tokenization

- This tokenization task is possible by searching spaces between the words.
- The root words used for information retrieval applications
- **Input:** Marathi language sentence/ document
- **Output:** list of tokens
- Consider the input sentence as: मराठी भाषा हे महाराष्ट्राचे वैभव आहे.
- The tokenization for the given sentence is:
 - मराठी
 - भाषा
 - हे
 - महाराष्ट्राचे
 - वैभव
 - आहे

Stemmer

- Removes all possible suffixes
- Reduces the word to its stem.
- Uses suffix list to remove suffixes from words.
- **Problem of Stemming:** Stem contains inflections which cannot be removed using simple stemming operation.
- **Examples of Suffixes:**
- च्या, न्या, ज्या ज्या ख्या च चा ची चे ही नी ने ला, कडे, कडून , देखील, वर , खाली, पर्यंत, मागे मागेच , समोर, सुद्धा, साठी, मधे मधेच मधेही मधेपण मधेसुद्धा , विरुद्ध
- **Input:** List of words
- **Output:** Stem of words
- **Example:**
- **Input:** मराठी भाषा हे महाराष्ट्राचे वैभव आहे.
- **Output:** मराठी भाषा हे महाराष्ट्रा वैभव आहे.

Morphological Analyzer

- To recognize the inner structure of the word and to produce Root words for a given input document .
- Stemmed words are analyzed to check whether they are inflected or not.
- **Input:** i) **Stemmed token** (with inflection), ii) List of **inflection stripping rule** :
- डा, डी, डे, डो, डौ, ड्या - ड
- ता, ति, ती, तु, तू, ते, तै, तो, तौ, तं, तः, त्या - त
- ना, नि, नी, नु, नू, ने, नै, नो, नौ, नं, नां, नः - न
- **Output:** Root words
- **Example** : In the word “महाराष्ट्राचे ” the stem form is “महाराष्ट्रा ” (maharashtra) whereas the root form is “महाराष्ट्र”.
- **Input:** मराठी भाषा हे महाराष्ट्रा वैभव आहे.
- **Output:** मराठी भाषा हे महाराष्ट्र वैभव आहे.

Tag Generator

- Well-chosen tagset and also important to represents parts of speech

Universal tagset for all ILs

- 1. [N] Noun
- 2. [V] Verbs
- 3. [PR] Pronoun
- 4. [JJ] Adjectives
- 5. [RB] Adverb
- 6. [PL] Participles
- 7. [PP] Postpositions
- 8. [DM] Demonstratives
- 9. [QT] Quantifier
- 10. [RP] Particle
- 11. [PU] Punctuation
- 12. [RD] Residual

Tagset for Marathi

No	Name	Tag	Description	Example
1	NOUN	NN	Common Nouns	मुलगा, साखर, मंडळी, चांगुलपणा
		NNP	Proper Nouns	मोहन, राम, सुरेश
		ABN	Abstract noun	गर्व, कौशल्य, क्रोध, चपळाई, गोडवा
2	PRONOUN	PPN	Personal pronoun	मी, आम्ही, तुम्ही
		PPS	Possessive pronoun	माझा, माझी, तुझा, तुझी, त्याचा
		PDM	Demonstrative pronoun	तो, ती, ते, हा, ही
		PRF	Reflexive pronoun	आपण, आम्ही, तुम्ही, तुम्हाला
		PRC	Reciprocal pronoun	एकमेकांचा, एकमेकाला, आपल्याला
3	ADJECTIVE	JJ	Modifier of Noun	उत्साही, श्रेष्ठ, बळवान
4	VERB	VM	Verb Main	बसणे, दिसणे, लिहिणे, पडला, झोपला
		VAUX	Verb Auxiliary	नाही, नको, करणे, हवे, नये

Tagset for Marathi

5	ADVERB	RB	(Modifier of Verb)	आता, काल, कधी, नेहमी, लवकर, हळूहळू
6	CONJUNCTION	CC	Coordinating and Subordinating	आणि, पण, जर, तर
7	POSTPOSITION	PSP	Postposition	आणि, वर, कडे, जवळ
8	INTERJECTION	INJ	Interjection	आहा, छान, अगो, हाय
9	NUMERAL	NUM	Number	१, २, ३, ४
		NUMCD	Cardinal Numeral	एक, दोन, तीन
		NUMO	Ordinal Numeral	पहिला, दुसरा, तिसरा
10	RESIDUAL	RDS	Symbol residual	\$, &, *, (,)
		RD_PUNC	Punctuation	?, ;, : !

Tagset for Marathi

11	REDUPLICATION	RDP	Reduplications	जवळ-जवळ
12	NEGATIVE	NEG	Negative	नाही,नको
13	DETERMINER	QF	Quantifiers	किती,पुष्कळ,खूप,भरपूर,बरेच
14	QUESTION WORDS	WQ	Question Words	काय, कधी, कु ठे
15	INTENSIFIER	INTF	Intensifier	खूप, फार,बराच,अतिशय
16	PARTICLES	RP	Particles	तर,ओहो
17	PHRASE	PHR	Phrase	नमस्कार,अभिनंदन,खेद आहे
18	ECHO	ECH	Echo Word	जेवणबिचण,डोकेबिके,पडणबिडण
19	QUATATIVE	UT	Quatative word	म्हणजे

Algorithm

- **Input:** i) Root word ii) Corpus for POS tagging
 - **Output:** Tagged word
- 1) Take root word from morphological analyzer
 - 2) Store all root words into array WORD
 - 4) Select each word one by one from array WORD and Search and compare selected word from corpus for pos tagging.
 - 5) If word is found one or more times, then store associated tag or tags of word into array TAGS
 - 6) Else display “the word is not found” add this new word with corresponding tag into lexicon with linguistic rules for new word.
 - 7) If one tag is stored in array TAGS, then display word with associated tag as an output.
 - 8) Else apply rule to select most appropriate tag for word.
 - 9) Display word with associated tag as an output.
 - 10 End.

Example

Input	मराठी	भाषा	हे	महाराष्ट्राचे	वैभव	आहे	.
Output	मराठी	भाषा	हे	महाराष्ट्र	वैभव	आहे	.
Tag	NNP	NN	PDM	NNP	NNP/JJ	VAUX	RD_PUNC

Disambiguation

- The "correct" and the "incorrect" assignment of a tag in particular contexts.
- Disambiguation is also based on contextual information or word / tag sequences.
- The ambiguity which is identified in the POS tagging module is resolved using the grammar rules.

WSD Rules for Marathi

Sr. No.	Ambiguity Between	Previous <u>word</u>	Next word	Possible Tag
1	Adjective and Adverb	Noun	Noun or Adjective	Adjective
			Verb	Adverb
2	Noun and Adjective	Not Noun	Noun	Adjective
		Noun	Verb	
3	Noun and Verb	Noun	Verb	Adverb
4	Auxiliary Verb and Noun	Noun	Vaux main	Verb
5	Verb and Auxiliary Verb	Vaux main	Verb	Adverb
6	Conjunction Noun Postposition	Noun, Pronoun,	NULL	Postposition
		Verb		Conjunction
		< Tag not matched>	< Tag not matched>	Noun (Default)
7	Noun, Adjective, Verb	NULL	Noun, Adjective, Verb	Noun
			< Tag not matched>	Verb (Default)

Example

Input	मराठी	भाषा	हे	महाराष्ट्राचे	वैभव	आहे	.
Output	मराठी	भाषा	हे	महाराष्ट्र	वैभव	आहे	.
Tag	NNP	NN	PDM	NNP	JJ	VAUX	RD_PUNC

Ambiguity removed by using WSD rule

Sample output of Marathi POS Tagger

मराठी शब्द जाती

Marathi POS Tagger

*** मराठी शब्द जाती ***

INPUT

रामायण हे वाल्मीकी ऋषी रच एक महाकाव्य आहे . भारत रामायण खूप प्रसिद्ध आहे . रामायण हिंदू धर्मीय एक पवित्र ग्रंथ आहे . श्रीराम हे विष्णू रूप हो . भारत गुरु स्थान खूप श्रेष्ठ आहे . प्रत्येक आयुष्य गुरु ज्ञान खूप उपयोग ठर .

OUTPUT

रामायण::NNP हे::PDM वाल्मीकी::NNP ऋषी::NN रच::VM एक::NUMCD महाकाव्य::NN आहे::VAUX ::RD_PUNC
भारत::NNP रामायण::NNP खूप::INTF प्रसिद्ध::JJ आहे::VAUX ::RD_PUNC रामायण::NNP हिंदू::NN धर्मीय::NN एक::
NUMCD पवित्र::JJ ग्रंथ::NN आहे::VAUX ::RD_PUNC श्रीराम::NNP हे::PDM विष्णू::NNP रूप::NN हो::INJ ::
RD_PUNC भारत::NNP गुरु::NN स्थान::NN खूप::INTF श्रेष्ठ::JJ आहे::VAUX ::RD_PUNC प्रत्येक::QF आयुष्य::NN गुरु::
NN ज्ञान::NN खूप::QF उपयोग::NN ठर::VM ::RD_PUNC

Clear Screen

Brows File

Validation

Tokenization

Stemmer

Morphology

POS Tag

Disambiguation